

Problem Statement ID

26043

Problem Statement Title

A digital platform to crowdsource societal challenges and facilitate collaborative problem solving through universities and industry partnerships

Description

Background:

Communities across Jharkhand encounter numerous local challenges related to education, healthcare, agriculture, water management, sanitation, environment, rural livelihoods, accessibility, urban infrastructure, and public service delivery. While citizens are often the first to identify these issues, there is currently no structured mechanism through which they can submit such problems for systematic evaluation and innovation-driven resolution. At the same time, Higher Education Institutions (HEIs) possess significant academic expertise, research capabilities, and a large pool of students capable of developing practical solutions. Industries and start-ups also have technical expertise, financial resources, and implementation capabilities that can complement academic research. However, collaboration among citizens, universities, and industry remains largely fragmented and project-specific. The National Education Policy (NEP) 2020 emphasizes experiential learning, multidisciplinary research, innovation, industry collaboration, and community engagement. Establishing a technology-enabled platform that connects societal challenges with academic institutions and industry partners can foster demand-driven innovation while enabling students and researchers to work on real-world problems that create measurable social impact.

Description:

Every year, citizens across Jharkhand identify thousands of local issues that require innovative technological or process-based solutions. These challenges often remain unresolved due to the absence of a centralized platform that enables problem collection, categorization, expert evaluation, institutional assignment, and industry collaboration. There is a need to develop a digital platform capable of:

- Allowing citizens, community organizations, local bodies, and government agencies to submit societal challenges through an intuitive web and mobile interface, supported by photographs, videos, location details, and relevant documents.
- Automatically categorizing submitted problems based on thematic domains such as education, agriculture, healthcare, water resources, environment, energy, urban development, accessibility, public administration, and rural livelihoods using AI-enabled classification techniques.
- Routing validated problem statements to appropriate universities based on their academic disciplines, research expertise, innovation centres, incubation facilities, and faculty specialization.
- Enabling universities to evaluate submitted challenges, constitute multidisciplinary student and faculty teams, and prepare solution proposals or research projects.
- Facilitating collaboration between universities and industry partners, startups, MSMEs, CSR organizations, research laboratories, and innovation ecosystems for mentorship, funding, prototyping, testing and deployment of solutions.

- Providing workflow management for problem review, institutional allocation, project monitoring, stakeholder communication, milestone tracking, and solution validation.
- Generating dashboards and analytics for government departments to monitor the number of challenges received, domain-wise distribution, institutional participation, industry engagement, project progress, and measurable social outcomes.

The platform should support a transparent and scalable innovation ecosystem that transforms community-driven challenges into research, innovation, entrepreneurship, and deployable solutions.

Expected Solution:

A comprehensive Societal Innovation Collaboration Portal comprising the following components:

- A citizen engagement module enabling individuals, community groups, Panchayati Raj Institutions, Urban Local Bodies, and government departments to submit societal challenges with multimedia evidence, geographical location, and supporting information.
- An AI-enabled problem management module capable of automatically categorizing, prioritizing, deduplication, and routing validated challenges to appropriate universities based on subject expertise and institutional capabilities.
- A university collaboration module allowing Higher Education Institutions to review assigned challenges, form multidisciplinary project teams, assign faculty mentors, manage project workflows, and submit solution proposals.
- An industry partnership module facilitating participation by industries, startups, MSMEs, CSR organizations, research institutions, and innovation hubs for mentoring, co-development, funding, prototyping, pilot implementation, and technology transfer.
- A project lifecycle management system for monitoring milestones, deliverables, approvals, documentation, testing outcomes, intellectual property generation, and implementation status.
- A visual analytics dashboard providing real-time insights on challenge submissions, university participation, industry collaborations, thematic trends, project completion rates, innovation outcomes, patents, startups created, and community impact across districts and sectors.
- A notification and communication system enabling seamless interaction among citizens, universities, industry partners, mentors, and government departments throughout the project lifecycle.

Organization Government of Jharkhand

Department Department of Higher & Technical Education

Category Software

Theme Smart Education